

The Ephemeris

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SJAA Booth @ San Jose Convention Center Advanced Imaging Conference

From Bill O'Neil

We had a very good day at the Advanced Imaging Conference held at the SJ Convention Center on Saturday, October 17, 2015.



Photo: SJAA Booth manned by Dave Ittner, Glenn Newell with a booth set-up assist from Rob Chapman
Photo Credit: Susan O'Neil

We set up the SJAA Lunt on the West Terrace with a new Sky Raider Mallin-cam Solar Camera and sent the video signal over the Wi-Fi to the convention floor. Visitors could see the images on our new super-bright 19" Monitor at the SJAA booth.



Above Photo: Take a guess!
Photo Credit: Susan O'Neil

Glenn is our IT genius and got all the techie problems solved!! It was 'mostly cloudy' but visitors got a kick out of seeing the clouds make the orange Sun go in and out of view. Better than Halloween!

Photo to the Right: Bill O'Neil setting up the Club's solar Lunt & Mallincam video camera on an outside terrace at the SJ Convention Center.

Photo Credit: Susan O'Neil



Dec - Feb 2016 Events

Board & General Meetings

Saturday 12/19, 1/23, 2/20

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Solar Observing (locations differ)

Sunday 1/3, 2/7

Fix-It Day (2-4pm)

Sunday 1/3, 2/7, 3/6

Houge Park In-Town Star Party

Friday 12/18, 1/15, 1/29, 2/12, 2/26

Intro to the Night Sky Class (Houge)

Friday 12/18, 1/15, 1/29, 2/12, 2/26

RCDO Starry Nights Star Party

Saturday 12/5, 1/30, 2/27

Binocular Star Gazing

Program will restart late Spring 2016

Imaging SIG Mtg

Tuesday 12/15, 1/19, 2/16

Quick START (by appointment)

Friday 12/11, 2/13

Unless noted above, please refer to the SJAA Web page for specific event times & locations.

SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.

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SJAA SHOW & TELL September 26, 2015

Photos From Ed Wong



SJAA at Coyote Valley Harvest Feast

From Bill O'Neil

Nearly 2,000 visitors came to the Coyote Valley Harvest Feast on Saturday, Sept. 19, 2015. This is the first year that Coyote Valley is an official Open Space Preserve open to the public, at 550 Palm Ave., in Morgan Hill. (Not too far from RCDO).



Coyote Valley Canopies provided by OSA

Photo credit: Teri Rogoway

SJAA brought its information Booth and 8 solar telescopes to show the public. Thanks to Terry Kahl, Marion Barker, Paul Colby, Ed Wong, Gary Chock, Bill O'Neil and Glenn Newell for bringing out their solar equipment.



Visitors at the SJAA booth

Nate McKenna and Jack Strong from the Eastbay Astronomical Society also joined us. Glenn brought his Mallincam video camera setup and attached it to the SJAA Lunt, so visitors could see spectacular Solar Flares on a very bright, 19" LCD Daylight Monitor (and not have to queue up at the eyepiece).



Marion Barker, Paul Colby and young astronomer

Photo Credit: Teri Rogoway

The Coyote Valley Harvest Feast is a family event that featured live entertainment, petting zoo, hikes and games for the kids and, of course: Solar Astronomy! KGO-TV's Dan Ashley was the Master of Ceremonies and performed with his band "Push".



SJAA's Marion Barker, Gary Chock, Ed Wong and Terri Kahl

All Photos taken by Bill O'Neil unless noted otherwise.



Glenn Newell showing live video of solar flares on SJAA's new TV monitor



Eyepieces

From the point of view of an amateur astronomer

From Rich Neuschaefer
Ephemeris April 2013 Article

Editors Note: I was going back through old issues of the Ephemeris looking for articles and found one on eyepieces from Rich Neuschaefer. The article is republished below.
Piller

These days eyepieces come in two barrel sizes, 1.25" and 2". A 2" eyepiece isn't necessarily better than a 1.25" eyepiece. Usually 2" eyepieces are used when you want wide true field and sometimes both a wide true field and a wide apparent field. Some 1.25" eyepieces have inner and outer barrels. This lets them fit into either a 1.25" or 2" eyepiece holder. Often these eyepieces have a relatively wide upper body with wide apparent fields. They often have a built-in Barlow that helps the eyepiece have a long eye relief.

True field is the number of degrees or arc minutes of sky that can be seen with a given telescope and eyepiece combination.

Apparent field is how large the hole seems to be when looking through the eyepiece. An eyepiece with a narrow apparent field gives the impression of looking through a soda straw. An eyepiece with a wide apparent field is more like looking through a picture window.

Exit pupil is diameter the beam of light coming out of the eyepiece. You would think a large (7 mm) exit pupil would be best for viewing deep sky objects, however, some very experienced deep sky observers feel a 1.5 mm exit pupil gives the best contrast between a deep sky object and the background sky. When observing planets at high power some of us have issues with a very small exit pupil. The small diameter beam can be easily interrupted by floaters in our eye. This is where it helps to have a large aperture instrument because it will increase the diameter of the beam of light for a given magnification. Eyepiece Exit pupil = telescope aperture / magnification A 6" aperture (153mm) telescope running 300x will give a 0.5mm exit pupil. A 14" aperture (355.6mm) telescope running 300x will give a 1.18mm exit pupil.

When observing the planets or tight double stars I don't mind not seeing the full field of view because the objects of interest usually don't take up much of the field of view. When observing the moon I very much enjoy an eyepiece with a wide apparent field. This is also true when observing large open clusters or large emission nebulae. Zoom eyepieces sometimes are more convenient than carrying several fixed focal length eyepiece. However, many of them have an annoying feature, the apparent field changes as you change the amount of zoom. At high zoom the appar-

ent field can be very nice but when zoomed to the low end of the power range the apparent field shrinks dramatically.

Most of the better eyepieces are fully multicoated meaning that every air to glass surface has been multi-coated. A fully multi-coated eyepiece will tend to have better contrast than an eyepiece that is not fully multi-coated or just coated.

Eye relief is measured in millimeters describing how close you need to get your eye to the eyepiece to see the full field of view. Sometimes the eye relief stated would seem to give you plenty of room but some eyepiece designs have the eye lens set down into the upper body of the eyepiece making it difficult to get close enough to see the full field, especially if you wear eye glasses when observing. The diameter of the eye lens (the lens closest to your eye) can give some idea of the eye relief of the eyepiece. Usually the larger the eye lens the better the eye relief.

Eyepiece designs like orthos and plossls with a long focal length usually have good eye relief, but as you go to shorter focal lengths the eye relief can get very short. One way to have a higher magnification and retain good eye relief is to use a Barlow lens. Some of the newer eyepiece designs have a number of lens elements but with good glass and being fully multi-coated the amount of light loss and scatter is quite small.

Virtually all 1.25" and 2" eyepieces have barrels that are threaded on the inside to accept a filter and to reduce light scatter. The threads are almost always painted flat black.

I find it is always nice to look through an eyepiece before buying. One benefit of attending star parties is that you can often have a look through a number of different eyepiece designs. I've had very few problems with the public looking through my good eyepieces. Today's multi-coated lenses hold up very well and are easy to clean. It is good to resist the desire to take apart an eyepiece. If some of the lens elements fall out the chances of getting the lenses back in the right order and with the correct side up is highly unlikely. Most telescope dealers don't have diagrams of how the lens elements should go in the eyepiece. It is a good idea to put your eyepieces away, out of the reach of little hands. The only eyepiece maker that I know of who will fix one of their eyepieces is TeleVue.

Magnification = the focal length of the telescope / the focal length of the eyepiece. Eyepieces and other items are sometimes available in the SJAA "For Sale" section in the club's blog area.



Science

Cloud Hovering over Titan's South Pole

Cassini's camera had already imaged an impressive cloud hovering over Titan's south pole at an altitude of about 186 miles (300 kilometers). However, that cloud, first seen in 2012, turned out to be just the tip of the iceberg. A much more massive ice cloud system has now been found lower in the stratosphere, peaking at an altitude of about 124 miles (200 kilometers).

The new cloud was detected by Cassini's infrared instrument – the Composite Infrared Spectrometer, or CIRS – which obtains profiles of the atmosphere at invisible thermal wavelengths. The cloud has a low density, similar to Earth's fog but likely flat on top.

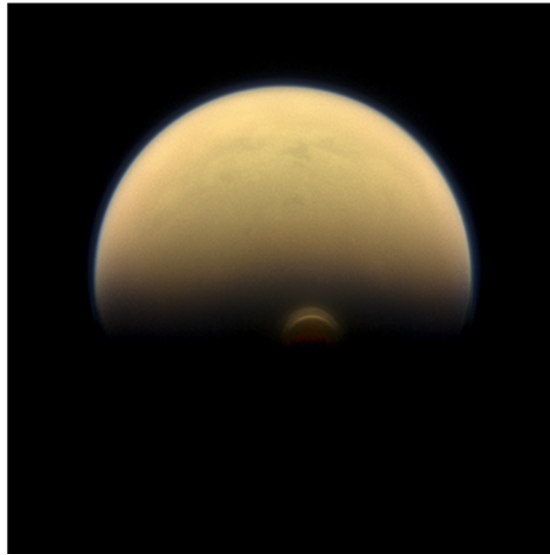
For the past few years, Cassini has been catching glimpses of the transition from fall to winter at Titan's south pole – the first time any spacecraft has seen the onset of a Titan winter. Because each Titan season lasts about 7-1/2 years on Earth's calendar, the south pole will still be enveloped in winter when the Cassini mission ends in 2017.

"When we looked at the infrared data, this ice cloud stood out like nothing we've ever seen before," said Carrie Anderson of NASA's Goddard Space Flight Center in Greenbelt, Maryland. "It practically smacked us in the face." Anderson is presenting the findings at the annual Meeting of the Division of Planetary Sciences of the American Astronomical Society at National Harbor, Maryland, on Nov. 11.

The ice clouds at Titan's pole don't form in the same way as Earth's familiar rain clouds. For rain clouds, water evaporates from the surface and encounters cooler temperatures as it rises through the troposphere. Clouds form when the water vapor reaches an altitude where the combination of temperature and air pressure is right for condensation. The methane clouds in Titan's troposphere form in a similar way. However, Titan's polar clouds form higher in the atmosphere by a different process. Circulation in the atmosphere transports gases from the pole in the warm hemisphere to the pole in the cold hemisphere. At the cold pole, the warm air sinks, almost like water draining out of a bathtub, in a process known as subsidence.

The sinking gases – a mixture of smog-like hydrocarbons and nitrogen-bearing chemicals called nitriles – encounter colder and colder temperatures on the way down. Different gases will condense at different temperatures, resulting in a layering of clouds over a range of altitudes.

Cassini arrived at Saturn in 2004 – mid-winter at Titan's north pole. As the north pole has been transitioning into springtime, the ice clouds there have been disappearing. Meanwhile, new clouds have been forming at the south pole. The build-up of these southern clouds indicates that the direction of Titan's global circulation is changing.



As winter sets in at Titan's south pole, a cloud system called the south polar vortex (small, bright "button") has been forming, as seen in this 2013 image.
Credits: NASA/JPL-Caltech/Space Science Institute

"Titan's seasonal changes continue to excite and surprise," said Scott Edgington, Cassini deputy project scientist at NASA's Jet Propulsion Laboratory (JPL) in Pasadena, California. "Cassini, with its very capable suite of instruments, will continue to periodically study how changes occur on Titan until its Solstice mission ends in 2017."

The size, altitude and composition of the polar ice clouds help scientists understand the nature and severity of Titan's winter. From the ice cloud seen earlier by Cassini's camera, scientists determined that temperatures at the south pole must get down to at least -238 degrees Fahrenheit (-150 degrees Celsius).



This 2012 close-up offers an early snapshot of the changes taking place at Titan's south pole. Cassini's camera spotted this impressive cloud hovering at an altitude of about 186 miles (300 kilometers). Cassini's thermal infrared instrument has now detected a massive ice cloud below it.

Credits: NASA/JPL-Caltech/Space Science Institute

The new cloud was found in the lower stratosphere, where temperatures are even colder. The ice particles are made up of a variety of compounds containing hydrogen, carbon and nitrogen.

Anderson and her colleagues had found the same signature in CIRS data from the north pole, but in that case, the signal was much weaker. The very strong signature of the south polar cloud supports the idea that the onset of winter is much harsher than the end.

"The opportunity to see the early stages of winter on Titan is very exciting," said Robert Samuelson, a Goddard researcher working with Anderson. "Everything we are finding at the south pole tells us that the onset of southern winter is much more severe than the late stages of Titan's northern winter."

The Cassini-Huygens mission is a cooperative project of NASA, ESA (European Space Agency) and the Italian Space Agency. JPL manages the mission for NASA's Science Mission Directorate in Washington. The CIRS team is based at Goddard.

For more information about Cassini, visit:

<http://www.nasa.gov/cassini>
<http://saturn.jpl.nasa.gov>

Credits: NASA/JPL-Caltech/Space Science Institute



Science

Out With the Old, In With the New: Telescope Mirrors Get New Shape

Telescope mirrors of old basically came in one shape: they were round and fit nicely inside a tube. No longer. An emerging optics technology now allows these light-gathering devices to take almost any shape, potentially providing improved image quality over a larger field of view — all in a smaller package.

Called freeform optics, this emerging mirror technology, brought about by advances in computer-controlled fabrication and testing, has triggered a sea change in optical engineering. Seeing the benefit of “potato chip-shape” or asymmetrical optics, NASA optical engineers at the Goddard Space Flight Center in Greenbelt, Maryland, have moved quickly to establish an expertise in this emerging technology.

“The use of freeform optics can significantly reduce the package size as well as improve the image quality,” said Joseph Howard, who is working with Goddard engineer Garrett West to ultimately design, integrate, and test a two-mirror freeform optical telescope for imaging and spectroscopic applications.

According to both Howard and West, the technology holds great promise for scientists who want to develop compact telescopes for CubeSat and other small satellites — an increasingly popular and cost-effective alternative to more traditional missions that are more expensive to build and launch.

“If you want to put these telescopes into a smaller box, you need to let the mirrors bend like a potato chip,” Howard explained.

With traditional two-mirror telescopes consisting of a primary light-gathering mirror and smaller, secondary mirrors, which relay the incoming light and direct it onto a detector, the rotationally symmetric — in other words, round — mirrors need to be aligned along the axis of the system to reduce optical aberrations that produce blurry images. With freeform optics, however, the asymmetric mirrors can better correct for these aberrations to provide a larger usable field of view, as well as dramatically reduce the light path, or package size.

As part of their research effort, Howard and West recently evaluated the optical system of a coastal measurement instrument, originally equipped with nine symmetrical mirrors. By replacing the mirrors with freeform optics, they

were able to reduce the size and number of mirrors to six, shrinking the telescope’s overall packaging by more than tenfold.

They also have selected a candidate two-mirror freeform optical telescope design, and are now awaiting the delivery of the two freeform mirrors with which they plan to assemble a prototype instrument for testing. “Our design studies suggest that a factor of five or more reduction in the volume of optical instrumentation can be achieved by freeform surfaces,” Howard said, adding that image quality also improves considerably.

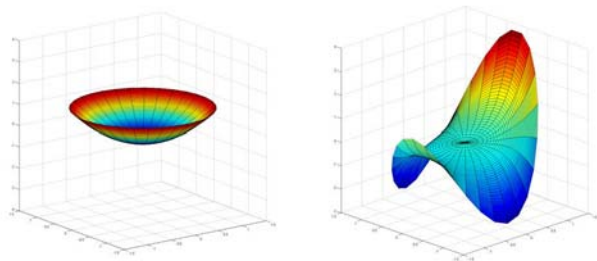
Next year, the team plans to continue testing its two-mirror instrument, which includes a freeform mirror manufactured with 3-D printing, also called additive manufacturing. This extends the work of another R&D effort that developed the first imaging telescopes assembled almost exclusively with 3-D manufactured parts. With this technique, a computer-controlled laser melts material in precise locations as indicated by a 3-D CAD model. Because

the mirror will be constructed layer by layer, it will be possible to construct a mirror with any shape.

The team believes the technology could prove to be game changing for a number of future missions, including instruments for imaging exoplanets. “NASA will benefit,” Howard said. “Freeform optics will be critical. They will enable larger fields of view and fit in size-limited packages, such as those found in CubeSats and small satellites, or on larger missions where space allocations are tight,” Howard said. To hasten the learning curve, Howard and his colleague, engineer Garrett West, established a group called the Freeform Optics Research Group Endeavor (FORGE). The group oversees freeform-optics research carried out by private industry under NASA’s Small Business Innovative Research program and Goddard scientists and engineers. The group already has implemented freeform-design practices in Goddard’s Optical Design Laboratory, known as the ODL, which provides design and engineering for instrument proposal efforts.

Other non-NASA research groups also are studying freeform optics, including the Center for Freeform Optics (CeFO), a National Science Foundation-sponsored cooperative research center headquartered at the University of Rochester in New York and the University of North Carolina-Charlotte. Currently, FORGE and CeFO are discussing possible ways to collaborate on instrument concepts using freeform optics, Howard said.

Credit: NASA’s Goddard Space Flight Center



A rotationally symmetric optic is traditionally used in telescopes. The freeform optic on the right takes a different shape and is now being investigated for use in space-based instruments.

Credits: NASA



Science

NASA Webb's Heart Survives Deep Freeze Test

After 116 days of being subjected to extremely frigid temperatures like that in space, the heart of the James Webb Space Telescope, the Integrated Science Instrument Module (ISIM) and its sensitive instruments, emerged unscathed from the thermal vacuum chamber at NASA's Goddard Space Flight Center in Greenbelt, Maryland.

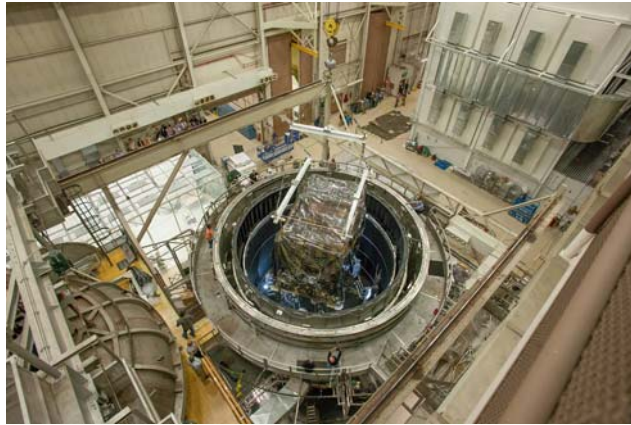
Teams of engineers and technicians have been on heart-monitoring duty around the clock since this complicated assembly was lowered into the chamber for its summer-long test.

Engineer Mike Drury, the ISIM Lead Integration and Test Engineer, is one of the test directors making sure that Webb will thrive in the frigid conditions at its final destination in space one million miles away from Earth. "The telescope is going to L2 or Lagrange Point 2, which is a very extreme environment," said Drury. "The heart of Webb called ISIM is a very important part of the observatory and will provide all of Webb's images."

These images will reveal the first galaxies forming 13.5 billion years ago. The telescope will also pierce through interstellar dust clouds to capture stars and planets forming in our own galaxy. Operating a telescope powerful enough to complete these tasks requires incredibly cold temperatures.

How cold? Try -387 degrees Fahrenheit, or 40 degrees Kelvin. This is 260 degrees Fahrenheit colder than any place on the Earth's surface has ever been. To create temperatures that cold on Earth, the team uses the massive thermal vacuum chamber at Goddard called the Space Environment Simulator, or SES, that duplicates the vacuum and extreme temperatures of space. This 40-foot-tall, 27-foot-diameter cylindrical chamber eliminates the tiniest trace of air with vacuum pumps and uses liquid nitrogen and even colder liquid helium to drop the temperature simulating the space environment.

"We complete these tests to make sure that when this telescope cools down, the four parts of the heart are still positioned meticulously so that when light enters the telescope we capture it the right way," said Paul Geithner, Webb's deputy project manager. "The biggest stress for this telescope will be when it cools down. When the telescope structure goes from room temperature to its super cold operating temperature, it will see more stress from shrinkage than it will from violent vibration during launch," said Geithner.



A crane lifts the heart of the James Webb Space Telescope from the Goddard Thermal Vacuum Chamber where it spent weeks in a space-like environment.

Credits: NASA/Desiree Stover

NASA photographer Desiree Stover captured the photo of ISIM as it was lowered into the chamber for testing. The heart of the telescope weighs about as much as an elephant. Inside its black composite frame the four science instruments are tightly packed and are specially designed to capture specific information about distant light in the universe.

"When I first started here at Goddard, the ISIM structure was completely bare," said Stover who has been at Goddard for two years. "Leading up to this test all four science instruments were integrated onto it, along with heat straps, harnesses and blankets."

Tightening the bolts and putting everything together beforehand required very dedicated teams. "When ISIM was lowered into the chamber at the start of the test, that was a pretty emotional moment that represented an intense amount of work," said Marc Sansebastian, a mechanical assembly, integration and test technician. "After ISIM traveled over-

head, we shifted back to technical mode because there are a million things that happen that you don't see."

At any given time of day during the test, the control room held representatives from all four-instrument teams. Each instrument has a test engineer, who makes sure the test is going well, and a data analyzer. Those teams are testing the hundreds of electrical connections and computer programs that give life to Webb's heart. "Kind of like having a car in a garage in the winter. You want to check the car to make sure that it is still working," said Alistair Glasse, instrument scientist for the Mid-Infrared Instrument (MIRI).

"The weather this year was phenomenal for the test. When the weather is bad, when it's humid and when it gets stormy that's when we run into problems with the chamber," said Ray Lundquist, ISIM Lead Systems Engineer. "At the beginning of the test, we had a couple of storms and the building got hit by lightning that shut the whole system down for 30 minutes, but since that storm we have had really great weather."

Once the test was completed, the team warmed up the chamber, and completed the final functional test

and a series of data analyses before they opened up the chamber.

"We've been very fortunate on this test. We've worked with all of the different teams. We have all been working shifts and pitching in," said Drury. "I'm really amazed at how well everyone is getting along together. We have a lot of people who are willing to help out."

The James Webb Space Telescope is the scientific successor to NASA's Hubble Space Telescope. It will be the most powerful space telescope ever built. Webb is an international project led by NASA with its partners, the European Space Agency and the Canadian Space Agency.

Learn more about the Webb telescope at:

www.jwst.nasa.gov
www.nasa.gov/webb

Credit: NASA's Goddard Space Flight Center, Greenbelt, Md.; Laura Betz



Science

Our Sun Came Late to the Milky Way's Star-Birth Party

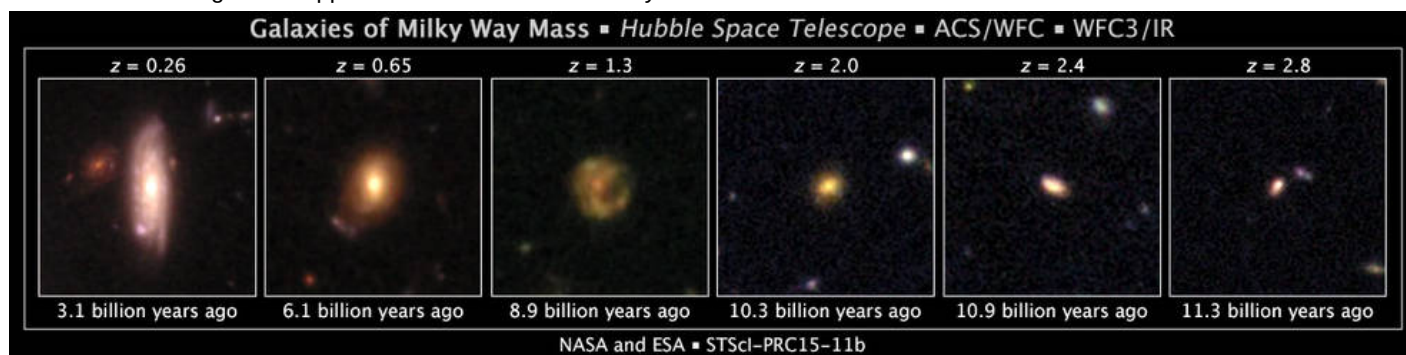
In one of the most comprehensive multi-observatory galaxy surveys yet, astronomers find that galaxies like our Milky Way underwent a stellar “baby boom,” churning out stars at a prodigious rate, about 30 times faster than today.

Our sun, however, is a late “boomer.” The Milky Way’s star-birthing frenzy peaked 10 billion years ago, but our sun was late for the party, not forming until roughly 5 billion years ago. By that time the star formation rate in our galaxy had plunged to a trickle. Missing the party, however, may not have been so bad. The sun’s late appearance may actually have fostered the growth of our solar system’s planets. Elements heavier than hydrogen and helium were more abundant later in the star-forming boom as more massive stars ended their lives early and enriched the galaxy with material that served as the building blocks of planets and even life on Earth.

Astronomers don’t have baby pictures of our Milky Way’s formative years to trace the history of stellar growth so they studied galaxies similar in mass to our Milky Way, found in deep surveys of the universe. The farther into the universe astronomers look, the further back in time they are seeing, because starlight from long ago is just arriving at Earth now. From those surveys, stretching back in time more than 10 billion years, researchers assembled an album of images containing nearly 2,000 snapshots of Milky Way-like galaxies.

The new census provides the most complete picture yet of how galaxies like the Milky Way grew over the past 10 billion years into today’s majestic spiral galaxies. The multi-wavelength study spans ultraviolet to far-infrared light, combining observations from NASA’s Hubble and Spitzer space telescopes, the European Space Agency’s Herschel Space Observatory, and ground-based telescopes, including the Magellan Baade Telescope at the Las Campanas Observatory in Chile.

“This study allows us to see what the Milky Way may have looked like in the past,” said Casey Papovich of Texas A&M University in College Station, lead author on the paper that describes the study’s results. “It shows that these galaxies underwent a big change in the mass of its stars over the past 10 billion years, bulking up by a factor of 10, which confirms theories about their growth. And most of that stellar-mass growth happened within the first 5 billion years of their birth.”



These six Hubble snapshots show how galaxies similar in mass to our Milky Way evolved over time. Milky Way-like galaxies grow larger in size and in stellar mass over billions of years.

Credits: NASA/ESA/C. Papovich (Texas A&M)/H. Ferguson (STScI)/S. Fabe

The new analysis reinforces earlier research which showed that Milky Way-like galaxies began as small clumps of stars. The galaxies swallowed large amounts of gas that ignited a firestorm of star birth.

The study reveals a strong correlation between the galaxies’ star formation and growth in stellar mass. So, when the galaxies slow down making stars, their growth decreases as well. “I think the evidence suggests that we can account for the majority of the buildup of a Milky Way-like galaxy through its star formation,” Papovich said. “When we calculate the star-formation rate of a Milky Way-like galaxy in the past and add up all the stars it would have produced, it is pretty consistent with the mass growth we expected. To me, that means we’re able to understand the growth of the ‘average’ galaxy with the mass of a Milky Way galaxy.”

The astronomers selected the Milky Way-like progenitors by sifting through more than 24,000 galaxies in the entire catalogs of the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey (CANDELS), taken with Hubble, and the FourStar Galaxy Evolution Survey (ZFOURGE), made with the Magellan telescope.

They used the ZFOURGE, CANDELS, and Spitzer near-infrared data to study the galaxy stellar masses. The Hubble images from the CANDELS survey also provided structural information about galaxy sizes and how they evolved. Far-infrared light observations from Spitzer and Herschel helped the astronomers trace the star-formation rate.

The team’s results will appear in the April 9 issue of The Astrophysical Journal.

For images and more information about the Hubble Space Telescope, visit: <http://www.nasa.gov/hubble> or <http://hubblesite.org/news/2015/11>

Credit: Space Telescope Science Institute, Baltimore, Maryland; Donna Weaver



SJAA Member Observation Posts

Pinnacles West Public Star Party

From Ed Wong

October 10th was our second public star party at the Pinnacles West Side. The comments from our Pinnacles Rangers and the public was that our program last night was a very resounding success and much appreciated. We had approximately 20-25 people from the public show up for the star party. Some heard about the program from the SJAA Meetup, some from the Pinnacles Facebook page or the visitors center. Most have never been to the Pinnacles West Side.



Photo: Pinnacles West line up (notice the AC paving; nice!)

Photo credit: Marilyn Perry

Thanks so much to the 5 astronomers we had at the Pinnacles West Side last night: Alexander Kramarov, Marilyn & Jonathan Perry, Gary Chock, Tom Piller, and Ed Wong. The night started out with our guests coming about 7:30pm, we started showing Saturn which was visible to the southwest to start with to entertain our guests. Ed did a greeting to our guests then did the "What's up in the night sky" talk. We then spent the next two hours showing our guests the night sky.

Objects that we notably commented on how spectacular they looked were: Saturn, M7, M8, M17, M22, M31, M32, M110, M57 (especially in

Alex's 12" Dob) Owl Cluster, Double Cluster, Albireo, Almach, Double Double. Our guests were also amazed to see Uranus and Neptune. One person said that seeing those 2 planets completed his bucket list of seeing all of the planets now.

After our guests left we spent time viewing and exploring on our own. One new object I was able to see for the first time was the Pacman Nebula. Marilyn was excited that she finally was able to find and see the Helix Nebula in her own scope. I spent sometime looking at the Double Cluster in my scope to discover many new details about it I don't remember being able to see before. The conditions were pretty good. The stars in the Double Cluster were so pinpoint sharp last night and the sky was dark, so there was good contrast. I have to say it might possibly be the best view I ever had of the Double Cluster. Of course me saying that sparked a comparison (known to happen amongst astronomers under dark skies) of who had the best view of the Double Cluster.

We were able to see Orion rising around midnight. It was a very memorable night for me. Great guests, great company to view with and great skies to view under.



Photo: Ed Wong and Gary Chock covering up the light sensor to keep the exterior lights from coming on during bathroom breaks.

Photo credit: Marilyn Perry



SJAA Member Observation Posts

Moon Occult of Aldebaran

From Mark McCarthy

I noticed in S&T's October issue (page 49) the moon was to occult of Aldebaran the morning of October 2, 2015. I checked the times and found they would fit nicely in my morning schedule. Disappearance was at 6:17am, reappearance at 7:17am. That would leave enough time to get up, observe the disappearance, have breakfast and get ready for work, observe the reappearance and leave for work right after. Luckily the sky was clear! I left my Astroscan in the kitchen overnight so I just needed to bring it outside to observe; and bring it in afterward.

I went outside shortly after 6:00am (still in my bathrobe!) and quickly found Aldebaran fairly wide from the bright side of the moon (@ 89x). It was nice to see Orion, the wintertime favorite, high in the sky with Canis Major prancing at his heels. I looked at the Orion Nebula; it was there, and I could see the Trapezium, but transparency was not very good with some thin fog in the air. Back to the moon; Aldebaran was now noticeably closer. I watched as the moon moved perceptibly eastward toward the star – which was interesting since in the eyepiece the whole scene drifted the opposite direction, to the west. It was heading toward Aldebaran at a slight angle, from the telescopic SW to the NE. The effect would have been more dramatic if I had a tracking mount and centered the view on Aldebaran. The moving moon recalled my experience last week seeing shadow and light changes along the ter-

minator; it's wondrous to see change happen in the sky.

Now the moon's limb was nearly touching the star; it seemed to float inside the narrow, slightly out of focus band of light my telescope presented of the limb. Then the star winked out. It was so sudden! And there was no bringing it back, no matter how I moved the scope around, looking for the lost star.

I came back outside at 7:10 to find my eyepiece had fogged over – I forgot, it's now the time of year when we have to manage our equipment for dew! I used a different eyepiece with a wider view (38x) and could see the whole moon. The sky was robin's egg blue, no stars were visible. I stared at the moon's nocturnal area and waited, waited. I checked my watch—I was nervous I had missed it—then peered back in the eyepiece: Aldebaran was back in view. In the two seconds I looked at my watch, I had missed the instant of reappearance! The lesson is, be patient. Aldebaran was just beyond the edge of the nocturnal part of the moon, and had followed that south-westerly angle I noticed just before it disappeared.

This was my first lunar occultation of a star, and it was fun. I now have another kind of observing project to add to my backyard observing routine: double stars, planets, asteroids, the antics of the Galilean moons, shadows along the lunar terminator, and now lunar occultation. It would be fun to see a star's disappearance and reappearance under high power, especially if it happens near mountains on the moon's limb: I imagine it would be like watching a sun set and rise on earth.

A list of 2015 bright star occultation's is at: <http://www.lunar-occultations.com/iota/bstar/bstar.htm>

To Scale: The Solar System

For those of you who missed it, on October 2nd Gary Chock sent out a link to a video clip of the Solar System to scale that was really fun to watch. Link is listed below:

<https://youtu.be/zR3lqc3Rhfg>

Rob Jaworski, after viewing the video went on to comment:

"That was a pretty cool project these guys did. The size of the sun being in proportion to their model was interesting, unexpected, good job to them on that. Couple other things I noticed.

Distances in miles in their model, of each of these planets from their model sun:

Mars: 0.17
Jupiter: 0.57
Saturn: 1.1
Uranus 2.1
Neptune 3.5

There's almost a pattern of proportion here.

Jupiter is 3x the distance than Mars.

Saturn is 1.9x than Jupiter.

Uranus is 1.9x than Saturn

Neptune is 1.7x than Uranus.

The other thing I noticed: The ending credits are in the shape of the Mercury capsule, or other lunar mission reentry vehicle. Or am I making that up?

Rob (with too much caffeine)"



SJAA Member Observation Posts

Anyone out looking at the eclipse?

From Tom Piller

The wonders of the digital age sure do facilitate some interesting lines of communication! Below is an email string captured off of the SJAA Group Yahoo page, memorializing the **September 27, 2015 eclipse of the moon**. I tried to sort the emails out on the time line as they occurred. Hope I caught everybody. We can thank Ed Wong for starting the string. Looks like the farther north the observers were the better the viewing.



Sep 27, 2015 6:41

PM, Ed Wong - Anyone out looking at the eclipse? Skies clearing up on the west, the east still doesn't look that great.

Sep 27 Robert Spangler

Not looking good here in Gilroy. Don't know if I will be able to see through this stuff. It doesn't look thick but I'll see. Marine layer and clouds. :(Bob

Sep 27 Gary Chock

High clouds are clearing from west to east, but not looking good for San Jose.

Sept 27th Robert Trower

Not looking good in Hollister. Cloud cover to the east pretty much obscures everything. Sent from my iPad

Sep 27, 2015 7:37 PM, Marianne Damon

Yes, I'm at Foothill, parking lot 4, with a good-sized crowd. Some scopes, I'm in my "observing chair" with binoculars. Mood here is of fun & excitement. Vega visible, no moon yet. Some clouds with large clear patches. Sent from my iPhone

Sun, Sep 27, 2015 at 7:54 PM Robert Spangler

Subject: Anyone out looking at the eclipse?
Anyone having any luck. Watching Nasa live feed now and they have clouds too. :(Bob

Sun, Sep 27, 2015 at 7:57 PM, Tom Piller

Cloudy down south in Monterey county as well.

Sep 27, 2015, at 8:12 PM, Glenn Newell

Very faint glow through the clouds in Scotts Valley

Sep 27, 2015, at 8:26 PM, Marianne Damon

We've got a view now, here at Foothill.

Sep 27, 2015 8:31 PM, Saeko Reignierd

Now we see it also. We are at San Jose airport area.

Sent from my iPhone

Sun, Sep 27, 2015 at 8:50 PM, Andy Hull

Nice and clear from Fremont.

Sep 27, 2015, at 8:55 PM, Glenn Newell

I'm getting good stuff now in Scotts Valley.

Sep 27 Mark Toney

Took the Orion XT10 in search of low east horizons in Marin Headlands, but fog was cascading down the hills as

we approached. Ended up stopping in Larkspur Ferry parking lot and got great views starting at 7:20 as moon cleared low clouds to the east. Shared views with about 35 people who had gathered in small groups to look through the scope, mostly with 21mmEthos. Lots of fun as the moon changed colors and brightness during the next couple of hours.

Sunday, September 27, 2015, Andy Hull

Nice and clear from Fremont.

Sept 27 Partha Pratim Sanyal

Now when the show is over almost... The clouds go away! Able to watch the moon coming out of the darkness slowly at San Jose!

Sep 27 Sukhada Palav

I went up to Sierra Vista with some friends and we waited till it finally cleared up. Could see most of it...!

It was a nice night overall. No cold weather, could sit in a chair and enjoy the stars above (whatever was visible) along with views of San Jose down in the valley...

Sun, Sep 27, 2015 at 10:36 PM, Marilyn Perry

I waited for clouds to clear in Sunnyvale, and had my camera all ready to go. I got one photo of moon just coming out of totality at 8:23, then clouds closed in again, then a little glimpse and another photo at 8:29 shows a bright stripe on bottom left hand side, so the total eclipse was over. After I waited so long, I am very happy I got one photo, even if it is a little over exposed.



SJAA Member Astrophoto Gallery



Subject: LDN 1235 – The Dark Shark Nebula

From: Mark Striebeck

Location / date: Orange Spring Ranch, Mariposa, CA / September 17-18, 2015

OTA size / type: Takahashi TOA-130 mounted on Astro-Physics Mach 1 GTO

Camera: FLI MLx694 with focal reducer TOA 645 Super Reducer

Filters: Astrodon Luminance, Astrodon RGB

Integration: 5.4 hours; 50 darks, 20 flats, 50 bias

Software: Pixinsight 1.8



SJAA Member Astrophoto Gallery

Photo: Solar Flare

From: Bill O'Neil

September 19, 2015

Location: Coyote Valley OSA Preserve



Subject: M33

From: Kai Yang (First light with my new Takahashi-85)

Date: Sept/19/2015

Location: Parking lot where close to D.A.R.C Observatory

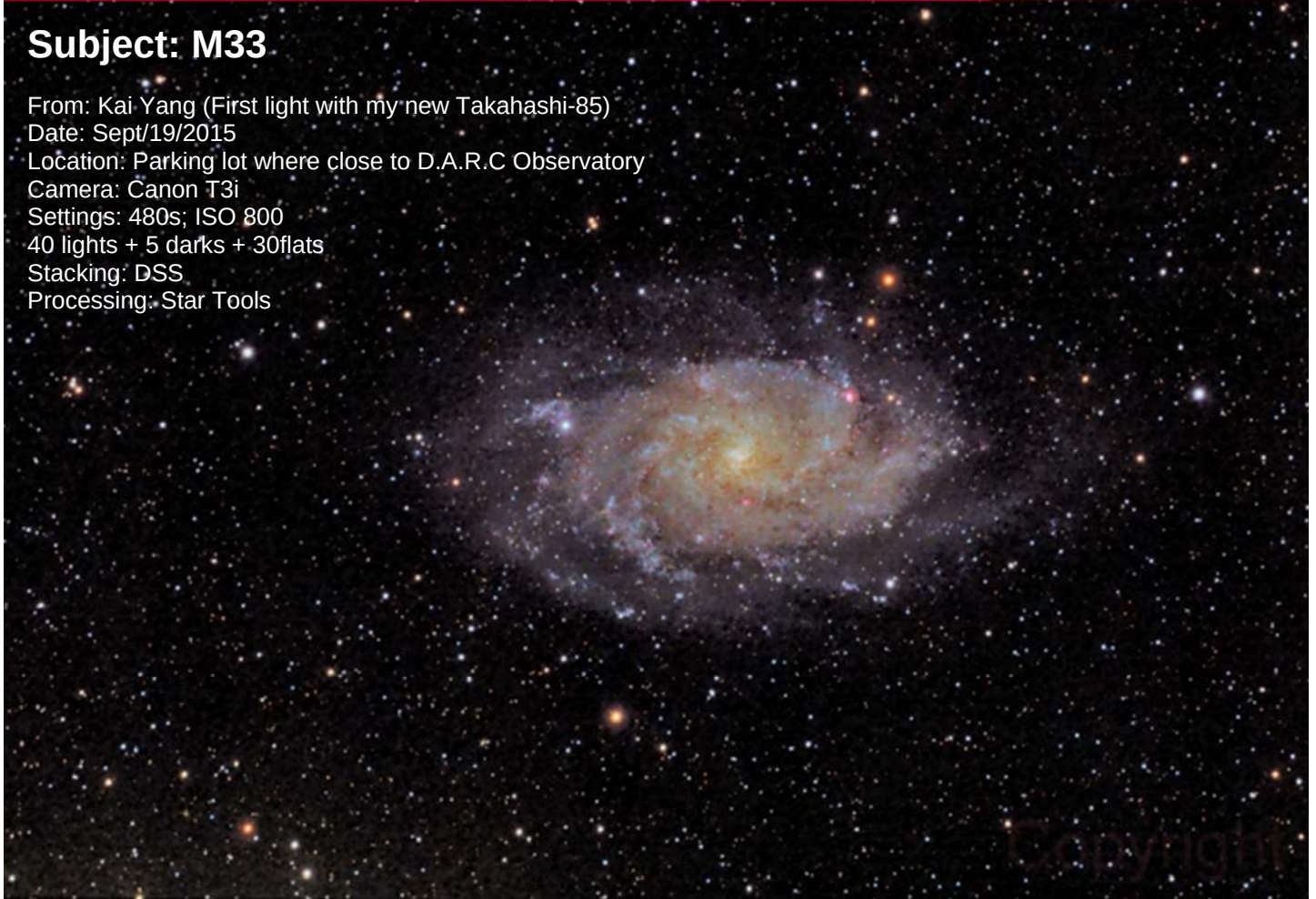
Camera: Canon T3i

Settings: 480s; ISO 800

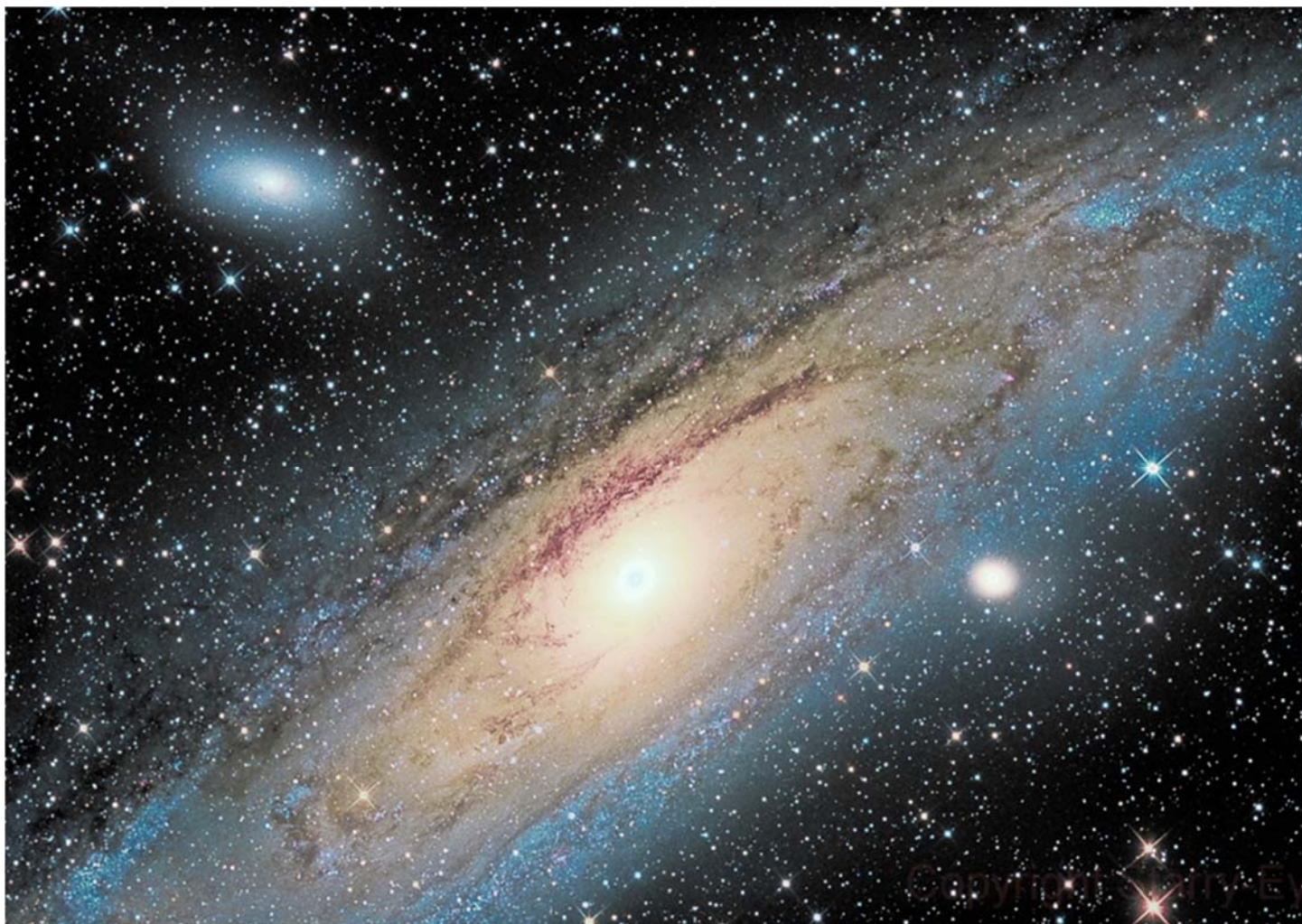
40 lights + 5 darks + 30 flats

Stacking: DSS

Processing: Star Tools



SJAA Member Astrophoto Gallery



Subject: M31

From: Paul Mahany

Location: Sierra Nevada, Yosemite, California

Imaging telescopes or lenses: ATM 6" F5 Newtonian

Imaging cameras: Canon 60Da

Mounts: Atlas EQ6 Atlas

Guiding telescopes or lenses: Orion 50mm guidescope Miniguider 50mm

Guiding cameras: Orion Starshoot Auto-Guider

Software: StarTools, DSS, BYEOS, SkyTools

Accessories: Paracorr II

Resolution: 2588x1728

Dates: Aug. 23, 2015

Frames: 95x420"

Integration: 11.1 hours

Avg. Moon age: 7.57 days

Avg. Moon phase: 52.03%

RA center: 10.512 degrees

DEC center: 41.204 degrees

Pixel scale: 2.163 arcsec/pixel

Orientation: 90.926 degrees

Field radius: 0.935 degrees

Smoky conditions in the Sierra's! Binned 2x2. Taken over 5 nights This August.



KID SPOT



Kid Spot Jokes:

- ♦ **When do astronauts have lunch?**
At launch time.
- ♦ **Which chocolate bars do you eat in space?**
Milky-ways and Mars bars.

Kid Spot Quiz:

What is Astronomy?

- A) The study of Physics
- B) The science of building space ships
- C) The study of outer space
- D) The study of telescopes

B. How many miles are in a light year?

- A) 6 trillion
- B) 93 million
- C) 10 million
- D) 3.6 miles

Did You Know?

Source: Astronomy Magazine March 2015 issue
"500 Coolest Things"

- ♦ In 1983 scientists verified that a group of 14 meteorites came from Mars.
- ♦ The Mariner 4 spacecraft performed the first successful flyby of Mars on July 14, 1965
- ♦ The Boomerang Nebula in the constellation Centaurus is the coldest natural place in the universe, with a temperature 1 kelvin above absolute zero.
- ♦ The largest crater in the solar system, Mars' North Polar Basin, covers 40 percent of the Red Planet.
- ♦ The first magnitude star Altair (Alpha Aquilae) rotates at a speed of 178 miles per second.
- ♦ According to NASA, the odds are some 250,000 to 1 that a falling meteorite will be the cause of your death.
- ♦ In 1655, Dutch astronomer Christian Huygens became the first to determine that Saturn has multiple rings.
- ♦ As your eyes adapt to the darkness, their sensitivity increases 10,000 fold in the first 30 minutes, with little gain after that.
- ♦ Lying 153 light years away the Hyades in the constellation Taurus is the nearest star cluster to Earth.

Kid Spot Quiz Answers:

C. The study of outer space.

A. 6 trillion



Club Updates

From the Board of Directors

Announcements

There will be an astro-imaging workshops held at Hogue Park Saturday January 30, 2016; 5:30-7:30pm and February 27, 2016; 6 - 8pm.

October 2015 Membership Report:
Total current membership: 367

Board Meeting Excerpts

August 29, 2015

In attendance

Dave Ittner, Ed Wong, Lee Hoglan, Greg Claytor, Bill O'Neil, Glenn Newell, Teruo Utsumi

Absent: (excused) Michael Packer (proxy to Lee)

Guests: Vini Carter, Ken Presti

SJAA 2016 Calendar

Dave produced the for 2016 calendar of events which was approved by all.

Mendoza Ranch

The club permit at Mendoza Ranch expires at the end of October 2015 and needs to be renewed

Initially only Saturday nights will be scheduled. Friday nights can be added if there are volunteers to become key holders.

SJAA partnership with Pinnacles

Ed is in contact with the US Forestry Service about potential use of the west side as an observing site. The planned pilot program is to host two star parties through the end of the year. The program will be re-evaluated after the second star party.

SJAA Survey

Dave's proposal to subscribe to Survey Monkey for one month to complete the SJAA survey was approved.

Video Astronomy

Glenn has performed a few events with his video gear, all successful. He proposed that SJAA purchase a monitor suitable for day time use at events. (Glenn is also willing to sell his video camera to the club). After discussion the Club decided to purchase a monitor (with the option to return if it isn't suitable for daytime use) for use at the OSA Coyote Valley event on 9/19 and consider the camera purchase at a later date.

Lunar eclipse

A viewing event for the upcoming lunar eclipse on 9/28 was mentioned. OSA is holding Sierra Vista, so OSA staff will not be available for any SJAA event. Discussion to continue on SJAA forum.

Board Seat

Greg Claytor submits his resignation to step down from his board seat. The SJAA thanks Greg for his years of service.

September 26, 2015

In attendance

Dave Ittner, Ed Wong, Lee Hoglan, Bill O'Neil, Glenn Newell, Teruo Utsumi
Absent: (excused) Rob Jaworski, Vini Carter

(unexcused) Michael Packer

Guests: Bruce Mitchell, Ida Mitchell, Gary Chock

Communications

Dave stressed the need for good communication among board members and asked everyone to do their part.

Member Participation

Dave lead the discussion on the need for more volunteers; the club has grown but the number of those willing to help run the club has not. As an example Ed is planning to transition from Mendoza Ranch duties to the Pinnacles and hence key holders for Mendoza are needed. Teruo will head up starting a task list to identify areas where help is needed.

Board Seat

Vini Carter has agreed to take the open board seat. Welcome aboard, Vini!

Video Astronomy

Bill purchased the daytime monitor. (Approved previous month w/ option to return.) It worked well at the OSA Coyote Valley day time event.

Hogue Park Time Conflict

The Youth Shakespeare Group has a set of events in the beginning of December and has a performance scheduled for Friday 12/4. To accommodate YSG the board decided to cancel the "Intro" class and in-town star part for that date.

New Checking Account

Rob opened a checking account at Key-point Credit Union to begin a transition from our current bank (Wells Fargo).

Mendoza Ranch Permit

Ed submitted the application and fee for the yearly Mendoza Ranch permit. Application is currently in processing.

Observing Event Request

The club received a request to do a viewing event at a Los Gatos winery for a customer. Bill to inquire further.

RASC order

Next year's RASC guide and calendar are coming out in a couple months. Teruo to facilitate pre-order form and announcement.

October 24, 2015

In attendance

Dave Ittner, Rob Jaworski, Ed Wong, Lee Hoglan, Bill O'Neil, Glenn Newell, Teruo Utsumi

Absent: (unexcused) Michael Packer

Guests: Ken Miura, Marianne Damon, Wolf Witt, Sukhada Palav

RASC 2016 Handbook and Calendar

The pre-order deadline for the RASC handbook will be extended.

Board of Directors Seat

The board discussed the lack of communication and participation from Michael Packer over the past several months. Attempts by the board, especially Dave, to contact Michael have had little response.

Glenn makes motion to remove Michael from the SJAA board. Ed seconds, approved by all.

Robert Armstrong Scholarship

Previous discussion on the scholarship was recapped. Rob will draft a proposal for the board to review next month.

Treasurer Needs

Rob explained the possible need to purchase a new version of Quickbooks and a stamp to facilitate processing checks.

Library

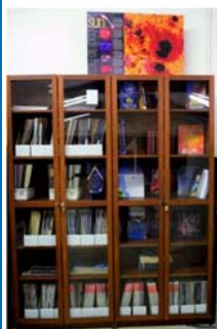
Sukhada expressed concern about the lack of accessibility to the library when those with the lock combination are not present. We also received a large donation and may need additional space. Also the bookshelf does not have sufficient depth for some books. It was noted that the magazines are rarely used. Sukhada to evaluate suggestions.

Solar Program

Bill has purchased the camera for SJAA use in the solar program: Mallicam Sky Raider SLP.



SJAA Library!



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astron-

mers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change:

<http://www.meetup.com/SJ-Astronomy/>

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally

held every other month. In addition to the indoor sessions, Ken Presti has offered to hold an outdoor session for the QS attendees to help them learn how to use the loaner scopes and the night sky.

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects. We'll need extra help to pull this off, but it appears to be doable."

The announcement and background can be found here:

<https://groups.google.com/forum/#!topic/sjaa-announcements/ScBqsdpNmzq>

Loaner Program

The purpose of this program, headed up by Manoj Koushik, is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment below. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

Astro Imaging Special Interest Group (SIG)

(SIG), led by Mark Striebeck, has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encour-

aged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups. Here's [the link](#).

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



SJAA Contacts

President:	Dave Ittner
Vice President:	Open
Treasurer:	Rob Jaworski
Secretary:	Teruo Utsumi
Director:	Lee Hoglan
Director:	Ed Wong
Director:	Bill O'Neil
Director:	Glenn Newell
Director:	Vini Carter
Memberships:	Ed Wong
Beginner Class:	vacant
Fix-it Program:	Vini Carter
Imaging SIG:	Mark Striebeck
Library:	Sukhada Palav
Loaner Program:	Manoj Koushik
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick STARt	Dave Ittner
Solar:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Teruo Utsumi
E-mails:	

<http://www.sjaa.net/contact>

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type:

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
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The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: sjaamemberships@gmail.com

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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Phone: _____

E-mail address: _____